

Proposal

ManipuriGPT

A Research-Grade Open Foundation Model for the Manipuri Language

1. Executive Summary

Manipuri (Meiteilon) is one of the officially recognized languages of India but remains significantly underrepresented in modern Large Language Models (LLMs). Existing multilingual models often demonstrate limited proficiency in Manipuri due to the scarcity of high-quality datasets, inconsistent script usage, and limited instruction-tuning resources.

This project proposes the development of **ManipuriGPT**, an open, reproducible, research-grade language model ecosystem designed specifically for Manipuri.

Rather than only fine-tuning an existing model, the project will build an entire ecosystem including:

- A unified multilingual corpus
- Multi-script understanding
- Language-adaptive pretraining
- Instruction tuning
- Retrieval-Augmented Generation (RAG)
- Open benchmark datasets
- Public evaluation suite
- Complete reproducible data pipeline

The objective is not merely to create another chatbot, but to establish an open foundation for future Manipuri NLP research.

2. Problem Statement

Current LLMs face several limitations when working with Manipuri:

- Very limited exposure during pretraining.
- Poor handling of Meitei Mayek.
- Weak support for Romanized Manipuri.
- Minimal understanding of historical Bengali-script Manipuri.
- Lack of high-quality instruction datasets.
- Very limited public benchmarks.

- Scarcity of openly available corpora.

As a result:

- Translation quality is inconsistent.
 - Reasoning capability is weak.
 - Cultural understanding is limited.
 - Educational usefulness is poor.
 - Domain-specific knowledge is almost nonexistent.
-

3. Vision

Develop the first open, research-grade Manipuri foundation model that understands the language rather than only individual scripts.

The model should naturally understand:

- Meitei Mayek
- Romanized Manipuri
- Bengali-script Manipuri

while generating responses primarily in modern Meitei Mayek (or another script if requested).

4. Objectives

Primary Objectives

- Build the largest reproducible Manipuri corpus.
 - Create a unified multilingual dataset.
 - Train a Manipuri-aware tokenizer.
 - Continue pretraining an existing LLM.
 - Fine-tune using instruction datasets.
 - Build RAG support.
 - Release everything publicly.
-

5. Core Principles

Reproducibility

Every experiment should be reproducible.

No manual downloading.

No hidden datasets.

Everything must be rebuildable.

Open Source

Datasets

Code

Models

Benchmarks

Documentation

Configuration Driven

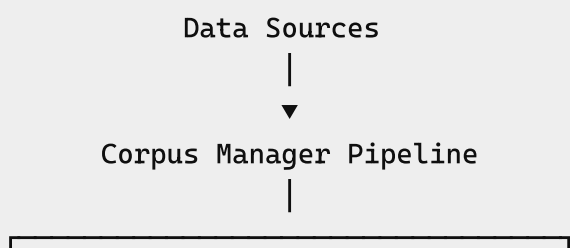
No dataset URLs should be hardcoded.

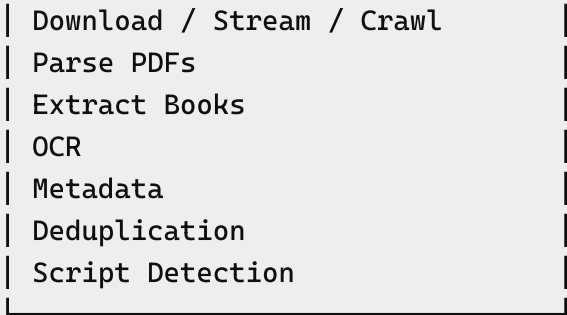
Everything will be managed through configuration files.

Modular

Every component should be replaceable independently.

6. Technical Architecture





Unified Corpus

Tokenizer Training

Language Adaptive Continued Pretraining

Instruction Fine-tuning (QLoRA)

DPO (Future)

Retrieval-Augmented Generation

ManipuriGPT

7. Multi-Script Strategy

The project will support three writing systems.

Meitei Mayek

Primary writing system.

Default output.

Roman Manipuri

Most common online.

Frequently used in chats and social media.

Bengali Script Manipuri

Historical script.

No longer dominant but extremely valuable for linguistic knowledge and legacy texts.

Instead of discarding Bengali-script data, it will be incorporated as an equivalent representation of the same language.

The model will learn semantic alignment across all three.

8. Corpus Categories

A. Parallel Corpora

Purpose:

Translation

Semantic alignment

Sources include:

- DayanandaThokchom English–Meitei
 - Ema-lon
 - NLLB
 - FLORES-200
 - AI4Bharat resources (where applicable)
-

B. Monolingual Corpora

Purpose:

Language adaptation

Sources include:

- MWireLabs Corpus
- Ema-lon Monolingual

- Wikipedia
 - CC-100
 - OSCAR
 - Common Crawl (filtered)
-

C. Educational Data

- Class VI–XII textbooks
 - Reference books
 - Notes
 - Question papers
-

D. Literature

- Stori
 - Nove
 - Poet
 - Folk tal
-

E. Government

- Notifications
 - Schemes
 - Acts
 - Circulars
-

F. Dictionaries

- Manipuri-English Dictionary
 - Lexicons
-

G. Grammar Resources

- Manipuri Grammar references
- Linguistic papers

Used for preprocessing and evaluation rather than language-model training.

H. OCR Resources

Future phase.

Includes handwritten character datasets.

9. Data Management Strategy

Datasets will not be manually collected into one folder.

Instead, every source will have metadata describing:

- Source
- License
- Script
- Domain
- Priority
- Download method
- Processing pipeline

Datasets hosted on Hugging Face will be streamed or cached through the `datasets` library rather than permanently stored.

Only resources such as PDFs, books, and websites requiring preprocessing will be downloaded locally.

10. Project Phases

Phase 0

Research

Corpus discovery

Dataset registry

Licensing review

Phase 1

Corpus ingestion

Streaming

Parsing

OCR

Website crawling

Phase 2

Cleaning

Unicode normalization

Deduplication

Script detection

Metadata generation

Phase 3

Tokenizer research

SentencePiece

Byte-Level BPE comparison

Vocabulary analysis

Phase 4

Language Adaptive Pretraining (DAPT)

Adapt a multilingual/open model to Manipuri using monolingual corpora.

Phase 5

Instruction Fine-tuning

QLoRA

Translation

Conversation

Question Answering

Summarization

Reasoning

Phase 6

Evaluation

BLEU

chrF

COMET

BERTScore

Human evaluation

Phase 7

Retrieval-Augmented Generation

Educational resources

Government documents

Reference books

Literature

Phase 8

Public Release

Dataset

Model

Tokenizer

Documentation

Benchmarks

Research paper

11. Compute Strategy

Local Machine (RTX 2060 6 GB)

- Development
 - Dataset ingestion
 - Cleaning
 - Tokenizer training
 - Evaluation
 - Inference
 - Small-scale experiments
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Google Colab Free

- QLoRA
 - Short DAPT experiments
 - Evaluation
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Kaggle Notebooks

- Longer free GPU sessions (subject to platform availability)
- Model experiments
- Dataset validation

The pipeline should be portable so that switching between local and cloud environments requires only configuration changes.

12. Technology Stack

Programming

- Python

Machine Learning

- PyTorch
- Transformers
- PEFT
- TRL
- bitsandbytes
- Accelerate
- Datasets
- SentencePiece

Vector Database

- FAISS

Data Processing

- Pandas
- Polars
- PyArrow

OCR

- Tesseract
- EasyOCR (future)

Experiment Tracking

- Weights & Biases (optional)
- TensorBoard

Deployment

- FastAPI
- Docker

13. Expected Deliverables

- Manipuri Corpus v1
- Multi-script corpus
- Dataset registry
- Corpus manager
- Tokenizer v1
- Continued-pretrained base model
- Instruction-tuned chat model
- Evaluation benchmark
- RAG knowledge base
- Public documentation
- Research publication draft

14. Success Criteria

The project will be considered successful if it produces:

- A reproducible end-to-end pipeline.
- An openly documented corpus.
- Improved Manipuri understanding compared with baseline multilingual models.
- Support for Meitei Mayek, Romanized Manipuri, and Bengali-script Manipuri.
- Publicly reusable datasets, models, and tooling that benefit future Manipuri NLP research. # ManipuriGPT

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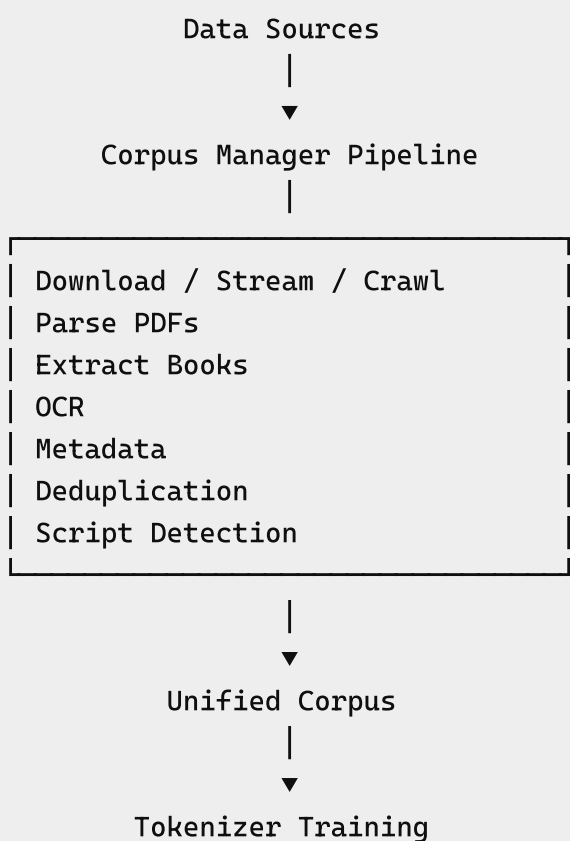
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